

ANANT GUPTA

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RESEARCH INTERESTS

Continual learning, meta-learning, and self-improving learning systems: how models acquire and retain knowledge over time without forgetting, and how to extract more training signal from limited data. I work mainly with language models (reinforcement learning, knowledge consolidation) and diffusion models.

EDUCATION

Georgia Institute of Technology

M.S. in Computer Science GPA: 4.0/4.0

Atlanta, GA

May 2027 (expected)

Georgia Institute of Technology

B.S. in Computer Science and B.S. in Mathematics GPA: 4.0/4.0

Atlanta, GA

May 2025

PUBLICATIONS

* denotes equal contribution.

Conference Papers

- [C1] **Trust Region Continual Learning as an Implicit Meta-Learner**
Z. Wang*, A. Gupta*, C.J. MacLellan
Conference on Neural Information Processing Systems (NeurIPS 2026). Poster. Atlanta, Dec. 2026.
- [C2] **CobwebTM: Probabilistic Concept Formation for Lifelong and Hierarchical Topic Modeling**
K. Singaravadivelan*, A. Gupta*, Z. Wang, C.J. MacLellan
Findings of the Association for Computational Linguistics (ACL 2026). Poster. San Diego, Jul. 2026.
- [C3] **Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion Models**
Z. Wang*, A. Gupta*, Z. Dong, C.J. MacLellan
International Conference on Learning Representations (ICLR 2026). Poster. Rio de Janeiro, Apr. 2026.
- [C4] **Hierarchical Semantic Retrieval with Cobweb**
A. Gupta*, K. Singaravadivelan*, Z. Wang
Advances in Cognitive Systems (ACS 2025). Oral Presentation. Atlanta, Oct. 2025.

Preprints and Papers Under Review

- [P1] **Cross-Task Transfer via Multi-Role Reinforcement Learning**
A. Gupta, Z. Wang, W. Xu, C.J. MacLellan
Under review, ICLR 2027.
- [P2] **Self-Consolidating Language Models: Continual Knowledge Incorporation from Context**
Z. Wang*, A. Gupta*, Z. Dong, C.J. MacLellan
Under review, ICLR 2027. [arXiv:2605.07076](https://arxiv.org/abs/2605.07076).
- [P3] **Test-Time Compositional Generalization in Diffusion Models via Concept Discovery**
Z. Wang, A. Gupta, T. Zhu, C.J. MacLellan
Under review, ICLR 2027. [arXiv:2605.07078](https://arxiv.org/abs/2605.07078).

Workshop Papers, Abstracts, and Presentations

- [W1] **LLMTutorBench: A Benchmark for University-level TCS AI Tutoring Systems**
A. Gupta*, H. Nguyen*, C.G. Webber*, J. Stevens, A. Ladha, S. Ainchwar, V. Ganesh
ACM SIGCSE Technical Symposium (SIGCSE TS 2026). Lightning Talk. St. Louis, Feb. 2026.
- [W2] **FinForge: Semi-Synthetic Financial Benchmark Generation**
G. Matlin, A. Theerthala*, A. Gupta*, A. JM, R. Castilla, Y.M. Ng, S. Chava
AAAI 2026 Workshop on AI for Finance. Poster. Singapore, Jan. 2026.

RESEARCH PROJECTS

Current

Role-Based Reinforcement Learning

Advisors: C. MacLellan, W. Xu

- Humans learn from the same material in more than one way, e.g., by comparing problems or identifying subgoals. Role-Based RL trains one language model in several roles over the same examples.
- Found cross-task transfer from the same data: auxiliary roles that never answer a question still improve problem solving, pointing to more training signal in fixed data than standard RL with verifiable rewards extracts [P1].
- Extending this to **meta-roles** for self-improvement, which decide which roles to create, which examples to view together, and how to allocate rollouts across roles during training.

LLMs as Debuggers in Unverifiable Domains

Advisor: V. Ganesh

- Building a benchmark of errors in physics papers from real open peer-review responses.
- Next, training challenger and grader roles through self-play, aiming for both a usable synthetic benchmark and a model that identifies errors where no verifiable reward exists.

Simulating Student Misconceptions

Advisor: V. Ganesh

- Studying whether LLMs can simulate a student in a single turn, producing a human-like incorrect answer conditioned on a given misconception (a test of counterfactual generation), and using these answers to build a synthetic benchmark for evaluating LLM graders in theoretical computer science [W1].

Completed

Continual Learning in Diffusion Models

Advisor: C. MacLellan

- Designed the experiments establishing that diffusion models admit an approximately rank-1 empirical Fisher, contributed to the proof, and ran the ImageNet experiments and qualitative analysis; the resulting Rank-1 EWC more than halves forgetting on ImageNet-1k relative to replay and diagonal-EWC baselines [C3].
- Contributed to the proof linking trust-region continual learning to MAML; ran the image-generation experiments, implemented baselines, and assisted with diffusion-policy experiments on Continual World [C1].

Continual Knowledge Consolidation in LLMs

Advisor: C. MacLellan

- SCoL uses meta-RL to teach an LLM which of its own layers to update when consolidating context [P2].
- Led the long-context study (LongBench v2), consolidating long passages under an intrinsic likelihood reward.
- Though meta-trained only on short contexts, SCoL outperforms in-context baselines that see the full passage and avoids the forgetting collapse of sequential fine-tuning on long passages.

Test-Time Compositional Generation

Advisor: C. MacLellan

- Contributed to the design of the method, which recovers concept prototypes from a single out-of-distribution query via score-based mode finding across noise levels of a pretrained diffusion model and composes them into a product-of-experts teacher; created the synthetic datasets used in the experiments [P3].

Hierarchical and Incremental Concept Formation

Advisor: C. MacLellan

- Formalized hierarchical semantic retrieval with Cobweb and implemented most of the system; it matches dense retrieval on strong encoders and stays robust where kNN collapses [C4].
- Proposed CobwebTM, a lifelong hierarchical topic model; designed, implemented, and ran its hierarchical topic modeling study, where it achieves the best topic and parent-child coherence against all baselines [C2].

TEACHING EXPERIENCE

Teaching Assistant, CS 4510: Automata and Complexity

Jan. 2024 – Dec. 2025

Georgia Institute of Technology

Head TA, Spring 2025

- Delivered two lectures to 300+ students, led 12 review sessions, and wrote homework and exam problems.

College of Computing Tutor

Jan. 2023 – Aug. 2023

Georgia Institute of Technology

- Tutored 150+ hours across AI, algorithms, data structures, and discrete mathematics.

SERVICE AND MENTORING

- **Reviewer**, International Conference on Learning Representations (ICLR 2027).
- **Reviewer**, NeurIPS 2026 Workshop on Towards Test-Time Continual Learning Agents (TTCL).
- **Research Mentor**, undergraduate researcher in Dr. Vijay Ganesh's Reasoning and Learning Group.